JAGANNATH UNIVERSITY

# PREVIOUS YEAR QUESTION BANK

B.Sc. (Honours) 4th Year 1st Semester – Theoretical Courses

DEPARTMENT OF BOTANY

**Compilation of Semester Final Questions from Batch 11 to Batch 16**
Academic Session: 2018-2019 to 2023-2024 (Exams held 2019-2024)
Based on the 4th Year 1st Semester Botany Curriculum Syllabi (BOT4101 - BOT4106)

***Compiled & Formatted programmatically by Sayad Hosine(17$^{th}$)***
**Published August 2026**

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# BOT4101: PHYSIOLOGY OF PLANTS GROWTH AND NUTRITION

## Syllabus Topic: Ion Absorption & Mineral Nutrition

**Batch: 11th Batch (2019) | Q. No: 1(a) | Marks: 2**

What is cation exchange theory of ion absorption?

**Batch: 11th Batch (2019) | Q. No: 1(b) | Marks: 5**

Describe chemiosmotic hypothesis of active absorption.

**Batch: 12th Batch (2020) | Q. No: 1(a) | Marks: 2.0**

Distinguish between active and passive absorption.

**Batch: 12th Batch (2020) | Q. No: 1(b) | Marks: 5.0**

With diagrammatic representation discuss Lundegardh's cytochrome pump theory.

**Batch: 12th Batch (2020) | Q. No: 7(a) | Marks: 3.5**

Write short note on: Chemiosmotic hypothesis

**Batch: 12th Batch (2020) | Q. No: 7(c) | Marks: 3.5**

Write short note on: Cation exchange theory

**Batch: 13th Batch (2021) | Q. No: 1(a) | Marks: 2.0**

What is Cation exchange theory of ion absorption?

**Batch: 13th Batch (2021) | Q. No: 1(b) | Marks: 5.0**

Describe Chemiosmotic hypothesis of active absorption.

**Batch: 14th Batch (2022) | Q. No: 2(a) | Marks: 6**

Explain Lundegardh's cytochrome pump theory for active ions absorption.

**Batch: 14th Batch (2022) | Q. No: 2(b) | Marks: 1**

Cite main pitfalls of this theory.

**Batch: 14th Batch (2022) | Q. No: 6(a) | Marks: 2**

Distinguish active absorption and passive absorption.

**Batch: 14th Batch (2022) | Q. No: 7(a) | Marks: 3.5**

Write short note on: Carrier concept theory

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 3**
Briefly explain carrier concept theory.

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 4**
Describe regulatory factors affecting the process of water absorption.

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 1**
Distinguish between active and passive absorption.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 6**
Explain chemiosmotic hypothesis.

**Batch: 16th Batch (2024) | Q. No: 2(a) | Marks: 2.0**
Mention the cation exchange theory of ion absorption.

**Batch: 16th Batch (2024) | Q. No: 2(b) | Marks: 5.0**
Describe Lundegarth's Cytochrome Pump theory for active ion absorption.

**Batch: 16th Batch (2024) | Q. No: 7(a) | Marks: 3.5**
Write short note on: Water absorption.

## Syllabus Topic: Translocation of Solutes

**Batch: 11th Batch (2019) | Q. No: 2(a) | Marks: 1**
What do you mean by translocation?

**Batch: 11th Batch (2019) | Q. No: 2(b) | Marks: 6**
Discuss different theories of mechanism of translocation.

**Batch: 11th Batch (2019) | Q. No: 7(a) | Marks: 3.5**
Write short note on: Phloem loading and unloading

**Batch: 12th Batch (2020) | Q. No: 2(a) | Marks: 2.0**
What is meant by phloem loading and unloading?

**Batch: 12th Batch (2020) | Q. No: 2(b) | Marks: 5.0**
Discuss on the path and mechanism of translocation.

**Batch: 13th Batch (2021) | Q. No: 2(a) | Marks: 1.0**

What is translocation?

**Batch: 13th Batch (2021) | Q. No: 2(b) | Marks: 6.0**

Discuss different theories of mechanism of translocation.

**Batch: 13th Batch (2021) | Q. No: 7(a) | Marks: 3.5**

Write short note on: Phloem loading and unloading

**Batch: 14th Batch (2022) | Q. No: 3(a) | Marks: 2**

What do you mean by source and sink?

**Batch: 14th Batch (2022) | Q. No: 3(b) | Marks: 5**

Discuss the mechanism of translocation.

**Batch: 14th Batch (2022) | Q. No: 6(b) | Marks: 3**

Distinguish between Phloem loading and Phloem unloading.

**Batch: 14th Batch (2022) | Q. No: 6(c) | Marks: 2**

Distinguish between Assimilates and Partitioning.

**Batch: 14th Batch (2022) | Q. No: 7(a) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write short note on: Assimilates and Partitioning

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 1.5**

Cite the name of translocated materials in phloem.

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 5.5**

Discuss Munch hypothesis with evidences and objections.

**Batch: 15th Batch (2023) | Q. No: 7(b) | Marks: 3.5**

Write short note on: Phloem loading and unloading

**Batch: 16th Batch (2024) | Q. No: 3(a) | Marks: 3.0**

What is phloem sap? Cite the name of translocated materials in phloem. (Marks: 1.0 + 2.0)

**Batch: 16th Batch (2024) | Q. No: 3(b) | Marks: 4.0**

Discuss on the rate and direction of phloem transport.

## Syllabus Topic: Nitrogen Metabolism

**Batch: 11th Batch (2019) | Q. No: 3(a) | Marks: 1**

Define nitrogen fixation.

**Batch: 11th Batch (2019) | Q. No: 3(b) | Marks: 6**

Discuss the process of formation of root nodule.

**Batch: 12th Batch (2020) | Q. No: 3(a) | Marks: 2.0**

What are the sources of nitrogen for plants?

**Batch: 12th Batch (2020) | Q. No: 3(b) | Marks: 5.0**

With suitable diagram describe the process of formation of root nodule.

**Batch: 12th Batch (2020) | Q. No: 6(a) | Marks: 5.0**

Briefly describe Burris hypothesis for biological nitrogen fixation.

**Batch: 13th Batch (2021) | Q. No: 3(a) | Marks: 1.0**

Define nitrogen fixation.

**Batch: 13th Batch (2021) | Q. No: 3(b) | Marks: 6.0**

With diagram describe the mechanism of biological nitrogen fixation.

**Batch: 14th Batch (2022) | Q. No: 4(a) | Marks: 2**

What are the sources of nitrogen for plants?

**Batch: 14th Batch (2022) | Q. No: 4(b) | Marks: 5**

With suitable diagram describe the process of formation of root nodule in Legume plants.

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 1**

Define nitrogen metabolism.

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 6**

Describe the mechanism of biological nitrogen fixation with labeled diagram.

**Batch: 16th Batch (2024) | Q. No: 4(a) | Marks: 2.0**

Write down the sources of nitrogen for plants.

**Batch: 16th Batch (2024) | Q. No: 4(b) | Marks: 5.0**

With diagram describe the process of formation of root nodule.

**Batch: 16th Batch (2024) | Q. No: 7(b) | Marks: 3.5**

Write short note on: Burris scheme of nitrogen fixation.

## Syllabus Topic: Plant Growth Regulators (PGRs)

**Batch: 11th Batch (2019) | Q. No: 4(a) | Marks: 2**

Describe the distribution of Auxins in plants.

**Batch: 11th Batch (2019) | Q. No: 4(b) | Marks: 5**

Mention physiological effects of Auxins.

**Batch: 11th Batch (2019) | Q. No: 5(a) | Marks: 4**

Briefly describe the role of growth hormones in fruit ripening.

**Batch: 11th Batch (2019) | Q. No: 5(b) | Marks: 3**

Mention different factors that affect fruit ripening.

**Batch: 11th Batch (2019) | Q. No: 7(c) | Marks: 3.5**

Write short note on: Gibberellins

**Batch: 12th Batch (2020) | Q. No: 4(a) | Marks: 3.0**

Define plant growth regulators and classify it with examples. (Marks: 1.0 + 2.0)

**Batch: 12th Batch (2020) | Q. No: 4(b) | Marks: 4.0**

Mention physiological effects of gibberellins on plant growth and developments.

**Batch: 12th Batch (2020) | Q. No: 7(b) | Marks: 3.5**

Write short note on: Cytokinins

**Batch: 13th Batch (2021) | Q. No: 4(a) | Marks: 1.0**

Define plant growth regulators.

**Batch: 13th Batch (2021) | Q. No: 4(b) | Marks: 6.0**

Explain physiological effects of auxins and cytokinins.

**Batch: 13th Batch (2021) | Q. No: 6(a) | Marks: 4.0**

Briefly describe the changes during fruit ripening.

**Batch: 13th Batch (2021) | Q. No: 6(b) | Marks: 3.0**

Mention different factors that affect fruit ripening.

**Batch: 14th Batch (2022) | Q. No: 5(a) | Marks: 3**

Classify plant growth regulators citing examples.

**Batch: 14th Batch (2022) | Q. No: 5(b) | Marks: 4**

Mention physiological effects of gibberellins on plant growth and developments.

**Batch: 14th Batch (2022) | Q. No: 7(c) | Marks: 3.5**

Write short note on: Fruit development

**Batch: 15th Batch (2023) | Q. No: 6(a) | Marks: 3**

Mention the role of growth hormones in fruit ripening.

**Batch: 15th Batch (2023) | Q. No: 6(b) | Marks: 4**

Describe the biochemical changes during fruit development.

**Batch: 15th Batch (2023) | Q. No: 7(a) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write short note on: Fruit development

**Batch: 15th Batch (2023) | Q. No: 7(c) | Marks: 3.5**

Write short note on: Artificial fruit ripening

**Batch: 16th Batch (2024) | Q. No: 5(a) | Marks: 1.5**

Classify plant growth regulators with example.

**Batch: 16th Batch (2024) | Q. No: 5(b) | Marks: 5.5**

Discuss physiological effects of gibberellins on plants.

**Batch: 16th Batch (2024) | Q. No: 7(c) | Marks: 3.5**

Write short note on: Role of growth hormones in fruit ripening.

## Syllabus Topic: Growth & Development

**Batch: 11th Batch (2019) | Q. No: 7(b) | Marks: 3.5**

Write short note on: Growth curve

**Batch: 13th Batch (2021) | Q. No: 7(c) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write short note on: Growth curve

**Batch: 14th Batch (2022) | Q. No: 1(a) | Marks: 2**

What is meant by growth and development?

**Batch: 14th Batch (2022) | Q. No: 1(b) | Marks: 5**

Briefly discuss the phases of growth and growth curve. (Marks: 3+2)

**Batch: 16th Batch (2024) | Q. No: 1(a) | Marks: 1.5**

Define differentiation, dedifferentiation & redifferentiation.

**Batch: 16th Batch (2024) | Q. No: 1(b) | Marks: 1.5**

Distinguish between growth and development.

**Batch: 16th Batch (2024) | Q. No: 1(c) | Marks: 4.0**

Briefly explain three distinct phases of Sigmoid curve.

## Syllabus Topic: Seed Physiology

**Batch: 11th Batch (2019) | Q. No: 6(a) | Marks: 3.5**

Define seed dormancy. Explain various causes of seed dormancy. (Marks: 1.0 + 2.5)

**Batch: 11th Batch (2019) | Q. No: 6(b) | Marks: 3.5**

Mention methods of breaking of seed dormancy.

**Batch: 12th Batch (2020) | Q. No: 5(a) | Marks: 2.0**

What do you mean by seed viability and seed dormancy?

**Batch: 12th Batch (2020) | Q. No: 5(b) | Marks: 5.0**

Discuss different morphological and biochemical changes during seed germination.

**Batch: 13th Batch (2021) | Q. No: 5(a) | Marks: 4.0**

Explain various causes of seed dormancy.

**Batch: 13th Batch (2021) | Q. No: 5(b) | Marks: 3.0**

Write down artificial methods of breaking seed dormancy.

**Batch: 14th Batch (2022) | Q. No: 7(b) | Marks: 3.5**

Write short note on: Methods of breaking dormancy

**Batch: 15th Batch (2023) | Q. No: 5(a) | Marks: 3**

Write about various causes of seed dormancy.

**Batch: 15th Batch (2023) | Q. No: 5(b) | Marks: 4**

Discuss about different morphological and biochemical changes during seed germination.

**Batch: 16th Batch (2024) | Q. No: 6(a) | Marks: 1.0**

What is seed germination & dormancy?

**Batch: 16th Batch (2024) | Q. No: 6(b) | Marks: 2.5**

Mention different types of seed dormancy.

**Batch: 16th Batch (2024) | Q. No: 6(c) | Marks: 3.5**

Discuss major way of breaking seed dormancy.

# BOT4102: NUMERICAL CYTOGENETICS

## Syllabus Topic: Numerical Chromosomal Aberrations

**Batch: 11th Batch (2019) | Q. No: 1(a) | Marks: 4**

Define numerical chromosomal aberration. How does numerical chromosomal aberration arise in plants? (Marks: 1+3)

**Batch: 11th Batch (2019) | Q. No: 1(b) | Marks: 3**

Give a brief overview on numerical chromosomal aberration.

**Batch: 11th Batch (2019) | Q. No: 7(a) | Marks: 3.5**

Write short note on: Numerical chromosomal aberration

**Batch: 13th Batch (2021) | Q. No: 7(a) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write short note on: Numerical chromosomal aberration

**Batch: 14th Batch (2022) | Q. No: 1(a) | Marks: 3**

Define numerical chromosomal aberration. How does numerical chromosomal aberration arise in plants? (Marks: 1+2)

**Batch: 14th Batch (2022) | Q. No: 1(b) | Marks: 4**

Give a brief overview on numerical chromosomal aberration.

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 1**

Define numerical chromosomal aberrations.

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 3**

With proper symbol summarize different types of aneuploid numerical chromosomal aberration.

**Batch: 15th Batch (2023) | Q. No: 1(c) | Marks: 3**

Narrate the major mechanisms responsible for numerical chromosomal abnormalities.

**Batch: 16th Batch (2024) | Q. No: 1(a) | Marks: 1**

What do you mean by numerical chromosomal aberrations?

**Batch: 16th Batch (2024) | Q. No: 1(b) | Marks: 3**

Give a list of common types of numerical chromosomal aberrations found in plants.

**Batch: 16th Batch (2024) | Q. No: 1(c) | Marks: 3**

Write down the causes and consequences of numerical chromosomal aberration.

### Syllabus Topic: Trisomics

**Batch: 11th Batch (2019) | Q. No: 2(a) | Marks: 1**

What do you mean by trisomic?

**Batch: 11th Batch (2019) | Q. No: 2(b) | Marks: 3**

Describe possible meiotic configurations found in primary and secondary trisomics for trivalent.

**Batch: 11th Batch (2019) | Q. No: 2(c) | Marks: 3**

Illustrate the segregation and transmission pattern of primary trisomics in plant.

**Batch: 12th Batch (2020) | Q. No: 1(a) | Marks: 1.0**

Define trisomy.

**Batch: 12th Batch (2020) | Q. No: 1(b) | Marks: 3.0**

Describe the types of trisomy based on the nature of the extra chromosome.

**Batch: 12th Batch (2020) | Q. No: 1(c) | Marks: 3.0**

Write down the origin and sources of primary trisomy.

**Batch: 13th Batch (2021) | Q. No: 1(a) | Marks: 2.0**

Write the classification of trisomic.

**Batch: 13th Batch (2021) | Q. No: 1(b) | Marks: 5.0**

With the help of checker board describe under what condition 35:1 and 8:1 ratio are observed in an offspring of primary trisomic.

**Batch: 13th Batch (2021) | Q. No: 2(b)(ii) | Marks: 1.5**

Write differences between: Trisomy and Triploid.

**Batch: 14th Batch (2022) | Q. No: 2(a) | Marks: 2**

Write the definition and origin of trisomy.

**Batch: 14th Batch (2022) | Q. No: 2(b) | Marks: 5**

With diagram show the configuration of primary, secondary and tertiary trisomics at M-I.

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 2**

What do you mean by hypoploidy and hyperploidy?

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 5**

Discuss genetic segregation ratio and transmission pattern of duplex trisomy.

**Batch: 16th Batch (2024) | Q. No: 2(a) | Marks: 3**

How does nondisjunction during meiosis contribute to hyperploidy and hypoploidy?

**Batch: 16th Batch (2024) | Q. No: 2(b) | Marks: 4**

How hyperploidy can be detected by expression, segregation, and the genetic ratio of a particular phenotype.

## Syllabus Topic: Monosomics & Nullisomics

**Batch: 11th Batch (2019) | Q. No: 3(a) | Marks: 1**

Define monosomic.

**Batch: 11th Batch (2019) | Q. No: 3(b) | Marks: 4**

Briefly give an exemplary note on the source of monosomics.

**Batch: 11th Batch (2019) | Q. No: 3(c) | Marks: 2**

Explain the meiotic and breeding behavior of monosomics.

**Batch: 11th Batch (2019) | Q. No: 7(a) | Marks: 3.5**

Write brief Notes on: Nullisomy

**Batch: 12th Batch (2020) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Monosomy

**Batch: 13th Batch (2021) | Q. No: 2(a) | Marks: 4.0**

How the position of genes are determined with the help of nullisomic?

**Batch: 13th Batch (2021) | Q. No: 2(b)(i) | Marks: 1.5**

Write differences between: Nullisomy and double monosomy.

**Batch: 13th Batch (2021) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Artificial production of monosomy.

**Batch: 14th Batch (2022) | Q. No: 3(a) | Marks: 2**

What are the types of monosomy?

**Batch: 14th Batch (2022) | Q. No: 3(b) | Marks: 5**

How the positions of dominant mutant genes are determined with the help of nullisomy?

**Batch: 15th Batch (2023) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Monosomy

**Batch: 16th Batch (2024) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Nullisomy

## Syllabus Topic: Haploids

**Batch: 11th Batch (2019) | Q. No: 4(a) | Marks: 4**

With example mention the natural source of haploids in flowering plants.

**Batch: 11th Batch (2019) | Q. No: 4(b) | Marks: 3**

Illustrate the importance of haploid in plant breeding.

**Batch: 12th Batch (2020) | Q. No: 3(a) | Marks: 1.0**

What is haploid?

**Batch: 12th Batch (2020) | Q. No: 3(b) | Marks: 4.0**

Describe the origin, occurrence and production of haploids.

**Batch: 12th Batch (2020) | Q. No: 3(c) | Marks: 2.0**

Mention the challenges and limitations to the use of haploids in plant breeding.

**Batch: 13th Batch (2021) | Q. No: 3(a) | Marks: 4.5**

Define haploid. Classify haploid in all possible ways.

**Batch: 13th Batch (2021) | Q. No: 3(b) | Marks: 2.5**

Illustrate the importance of haploid in plant breeding.

**Batch: 16th Batch (2024) | Q. No: 3(a) | Marks: 4.5**

Define haploid. Classify haploids in all possible ways.

**Batch: 16th Batch (2024) | Q. No: 3(b) | Marks: 2.5**

Illustrate the importance of haploids in plant breeding.

## Syllabus Topic: Polyploids & Meiotic Behavior

**Batch: 11th Batch (2019) | Q. No: 5(a) | Marks: 3**

Explain the phenotypic consequence of polyploids.

**Batch: 11th Batch (2019) | Q. No: 5(b) | Marks: 4**

Discuss the advantage and disadvantage of polyploids in agriculture.

**Batch: 11th Batch (2019) | Q. No: 7(b) | Marks: 3.5**

Write brief Notes on: Triploid

**Batch: 12th Batch (2020) | Q. No: 2(b) | Marks: 2.0**

Write the significance of polyploid in crop improvement.

**Batch: 12th Batch (2020) | Q. No: 4(a) | Marks: 1.0**

What is triploid?

**Batch: 12th Batch (2020) | Q. No: 4(b) | Marks: 6.0**

Mention the origin, phenotypic characteristics and meiotic behavior of triploid.

**Batch: 12th Batch (2020) | Q. No: 5(a) | Marks: 1.0**

Discuss the basic features of autotetraploid in meiosis.

**Batch: 12th Batch (2020) | Q. No: 5(b) | Marks: 6.0**

Depending on the number and orientation of centromere and the position of chiasma, show the configuration of autotetraploid chromosome in meiosis.

**Batch: 13th Batch (2021) | Q. No: 4(a) | Marks: 3.0**

Explain the chromatid level segregation of a triplex autotetraploid.

**Batch: 13th Batch (2021) | Q. No: 4(b) | Marks: 4.0**

With diagram show the identifying configurations of autotetraploid in meiosis.

**Batch: 14th Batch (2022) | Q. No: 4 | Marks: 7**

Briefly explain the synthetic classification of polyploid according to G. L. Stebbins.

**Batch: 14th Batch (2022) | Q. No: 5(a) | Marks: 3**
Describe the occurrence of 'X', 'Y' and 'Inter-locked' shaped quadrivalent at M-I of autotetraploid.

**Batch: 14th Batch (2022) | Q. No: 5(b) | Marks: 1**
Mention the factors affecting the meiotic orientation of autotetraploid.

**Batch: 14th Batch (2022) | Q. No: 5(c) | Marks: 3**
With the help of checkerboard explain chromosomal level segregation of a triplex autotetraploid.

**Batch: 14th Batch (2022) | Q. No: 7(a) | Marks: 3.5**
Write notes on: Triploid

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 2**
Distinguish between euploid and polyploid cell.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 3**
Interpret the consequences of polyploid in plant diversification, adaptation and evolution.

**Batch: 15th Batch (2023) | Q. No: 3(c) | Marks: 2**
Briefly classify polyploid with symbol.

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 1**
How do you define auto-tetraploid?

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 6**
Give an outline of all possible M-I configuration in auto-tetraploid plants.

**Batch: 15th Batch (2023) | Q. No: 7(b) | Marks: 3.5**
Write notes on: Triploid

**Batch: 16th Batch (2024) | Q. No: 4(a) | Marks: 1**
What is triploid?

**Batch: 16th Batch (2024) | Q. No: 4(b) | Marks: 6**
Mention the origin, phenotypic characteristics, and meiotic behavior of the triploid. (Marks: 2+2+2)

**Batch: 16th Batch (2024) | Q. No: 5(b) | Marks: 5**
Give an outline of all possible M-I configurations in autotetraploid plants.

## Syllabus Topic: Allopolyploidy, Segmental Allopolyploidy & Speciation

**Batch: 11th Batch (2019) | Q. No: 6(a) | Marks: 4**
With suitable example describe the application of allopolyploid in the origin of new species.

**Batch: 11th Batch (2019) | Q. No: 6(b) | Marks: 3**
"Origin of Raphanobrassica was economically a great success" – Discuss.

**Batch: 11th Batch (2019) | Q. No: 7(c) | Marks: 3.5**
Write brief Notes on: Segmental allopolyploid

**Batch: 12th Batch (2020) | Q. No: 2(a) | Marks: 5.0**
Explain with example "apparently auto-polyploid but actually allopolyploid."

**Batch: 12th Batch (2020) | Q. No: 6(a) | Marks: 1.0**
What is segmental allopolyploid?

**Batch: 12th Batch (2020) | Q. No: 6(b) | Marks: 6.0**
Give two examples of speciation through segmental allopolyploid.

**Batch: 12th Batch (2020) | Q. No: 7(b) | Marks: 3.5**
Write notes on: Triticale

**Batch: 13th Batch (2021) | Q. No: 5(a) | Marks: 4.5**
"Apparently autopolyploid but actually allopolyploid" – explain with example.

**Batch: 13th Batch (2021) | Q. No: 5(b) | Marks: 2.5**
Explain Winge and Digby hypothesis on the role of allopolyploid in speciation.

**Batch: 14th Batch (2022) | Q. No: 6(a) | Marks: 3**
With two suitable examples describe the role of allopolyploid in speciation.

**Batch: 14th Batch (2022) | Q. No: 7(c) | Marks: 3.5**
Write notes on: Segmental allopolyploid

**Batch: 15th Batch (2023) | Q. No: 5(a) | Marks: 1**
What is allopolyploidy?

**Batch: 15th Batch (2023) | Q. No: 5(b) | Marks: 6**
Discuss the role of allopolyploid in the origin of the genus Brassica and Triticale.

**Batch: 16th Batch (2024) | Q. No: 5(a) | Marks: 2**
What do you mean by autoallopolyploid and segmental allopolyploid?

**Batch: 16th Batch (2024) | Q. No: 6(a) | Marks: 1**
What do you mean by speciation?

**Batch: 16th Batch (2024) | Q. No: 6(b) | Marks: 6**
Discuss the role of allopolyploidy in Gossypium spp. And Brassica spp. (Marks: 3+3)

**Batch: 16th Batch (2024) | Q. No: 7(b) | Marks: 3.5**
Write notes on: Raphanobrassica

## Syllabus Topic: Mixoploidy & Human Cytogenetics

**Batch: 12th Batch (2020) | Q. No: 7(c) | Marks: 3.5**
Write notes on: Turner's syndrome

**Batch: 13th Batch (2021) | Q. No: 6(a) | Marks: 3.0**
What is gene balance theory and mixoploidy?

**Batch: 13th Batch (2021) | Q. No: 6(b) | Marks: 4.0**
Explain "Down" and "Turner" syndrome with symptoms.

**Batch: 14th Batch (2022) | Q. No: 6(b) | Marks: 4**
Explain 'Patau' and 'Turner' syndrome with symptoms.

**Batch: 14th Batch (2022) | Q. No: 7(b) | Marks: 3.5**
Write notes on: Mixoploidy and gene balance theory

**Batch: 15th Batch (2023) | Q. No: 6 | Marks: 7**
Narrate the causes, symptoms and consequences of Klinefelter syndrome, Edward syndrome and Triplo-X syndrome.

**Batch: 16th Batch (2024) | Q. No: 7(a) | Marks: 3.5**
Write notes on: Down's syndrome

## Syllabus Topic: Diploidization

**Batch: 13th Batch (2021) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Diploidization

**Batch: 15th Batch (2023) | Q. No: 7(a) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write notes on: Diploidization

# BOT4103: INTRODUCTION TO PLANT BIOTECHNOLOGY

## Syllabus Topic: Introduction & General Aspects

**Batch: 11th Batch (2019) | Q. No: 1(a) | Marks: 4**

Define biotechnology. Discuss on the historical background of plant biotechnology.

**Batch: 11th Batch (2019) | Q. No: 1(b) | Marks: 3**

With examples describe the importance of biotechnology in crop improvement.

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 3**

Explain the importance of plant tissue culture.

## Syllabus Topic: Laboratory Organization & Aseptic Techniques

*No final exam questions were asked on this specific topic from Batch 11 to 16.*

## Syllabus Topic: Culture Media & Stock Solutions

**Batch: 15th Batch (2023) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Components of MS medium

## Syllabus Topic: Micropropagation

**Batch: 11th Batch (2019) | Q. No: 5(a) | Marks: 4**

Define micropropagation. Mention the advantages and disadvantages of micropropagation. (Marks: 1+3)

**Batch: 11th Batch (2019) | Q. No: 5(b) | Marks: 3**

Write down the commercial exploitation of micropropagation with relevant examples.

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 4**

Write on the stages of micropropagation.

**Batch: 15th Batch (2023) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Stages of micropropagation

## Syllabus Topic: Organogenesis & Somatic Embryogenesis

**Batch: 11th Batch (2019) | Q. No: 4(a) | Marks: 3**

Define organogenesis. Mention the factors affecting organogenesis. (Marks: 1+2)

**Batch: 11th Batch (2019) | Q. No: 4(b) | Marks: 4**

Describe the modes of organogenesis for plant production.

**Batch: 11th Batch (2019) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Somatic embryogenesis

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 3.5**

Describe different stages of somatic embryogenesis.

### Syllabus Topic: Callus & Cell Suspension Cultures

**Batch: 11th Batch (2019) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Cell suspension culture

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 4**

Describe different phases of cell suspension culture.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 3**

Write down the advantages and disadvantages of cell suspension culture.

### Syllabus Topic: Protoplast Isolation, Culture & Fusion

**Batch: 11th Batch (2019) | Q. No: 3(a) | Marks: 5**

Briefly describe the methods of protoplast isolation.

**Batch: 11th Batch (2019) | Q. No: 3(b) | Marks: 2**

Compare between somatic hybridization and cybridization.

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 5**

Briefly describe the enzymatic method of protoplast isolation.

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 2**

Write down the importance of protoplast culture.

### Syllabus Topic: Haploid Production

**Batch: 11th Batch (2019) | Q. No: 2(a) | Marks: 4**

Discuss about the pathways of microspore division during androgenesis process.

**Batch: 11th Batch (2019) | Q. No: 2(b) | Marks: 3**

Mention the significance of androgenesis in crop improvement.

**Batch: 15th Batch (2023) | Q. No: 6(a) | Marks: 2**

How haploids are produced?

**Batch: 15th Batch (2023) | Q. No: 6(b) | Marks: 5**

Describe the process of haploid production through pollen culture.

## Syllabus Topic: Production of Virus-Free Plants

**Batch: 15th Batch (2023) | Q. No: 5 | Marks: 7**

Explain the production of virus free plants through heat therapy and meristem culture.

## Syllabus Topic: Cryopreservation & Germplasm Conservation

**Batch: 11th Batch (2019) | Q. No: 6(a) | Marks: 4**

What is cryopreservation? Discuss different steps of germplasm cryopreservation.

**Batch: 11th Batch (2019) | Q. No: 6(b) | Marks: 3**

Write the importance of germplasm conservation in crop improvement.

**Batch: 11th Batch (2019) | Q. No: 7(c) | Marks: 3.5**

Write notes on: In vitro conservation

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 3.5**

Briefly describe the method of cryopreservation.

**Batch: 15th Batch (2023) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Germplasm conservation

# BOT4104: ENVIRONMENTAL SCIENCE

## Syllabus Topic: Introduction & Key Themes

**Batch: 16th Batch (2024) | Q. No: 1(a) | Marks: 2**

Define environmental science. Mention the major themes of environmental science. (Marks: 1+1)

**Batch: 16th Batch (2024) | Q. No: 1(b) | Marks: 3**

Discuss about the main paradigm of environmental science and their key features.

**Batch: 16th Batch (2024) | Q. No: 1(c) | Marks: 2**

What are the essential intellectual standards of Science and values?

## Syllabus Topic: Environmental Health, Pollution & Toxicology

**Batch: 16th Batch (2024) | Q. No: 2(a) | Marks: 1.5**

Difference between pollution and synergism.

**Batch: 16th Batch (2024) | Q. No: 2(b) | Marks: 3.5**

With suitable example explain biomagnification.

**Batch: 16th Batch (2024) | Q. No: 3(a) | Marks: 4**

Define pollutants. Describe any three categories of pollutants. (Marks: 1+3)

**Batch: 16th Batch (2024) | Q. No: 3(b) | Marks: 3**

Briefly discuss about the ecological and biological effects of pollutants.

**Batch: 16th Batch (2024) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Persistent organic pollutants (POPs)

## Syllabus Topic: Atmosphere, Climate Change & Global Warming

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 4**

Explain the structure and importance of atmosphere.

**Batch: 15th Batch (2023) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Ozone layer

**Batch: 16th Batch (2024) | Q. No: 4(a) | Marks: 3**

Describe the structure of atmosphere.

**Batch: 16th Batch (2024) | Q. No: 4(b) | Marks: 2**

Discuss about the major greenhouse gases.

**Batch: 16th Batch (2024) | Q. No: 4(c) | Marks: 2**

Write about the environmental and ecological effects of global warming.

**Batch: 16th Batch (2024) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Ocean acidification

## Syllabus Topic: Energy, Resources & Sustainability

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 3**

Mention the six differences between renewable and non-renewable resources.

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 2.5**

Write about MIPS.

**Batch: 15th Batch (2023) | Q. No: 6(a) | Marks: 3**

Explain polluter pays principle.

**Batch: 15th Batch (2023) | Q. No: 6(b) | Marks: 4**

Mention the principles and applications of sustainability.

**Batch: 15th Batch (2023) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Exploitation of natural resources

**Batch: 16th Batch (2024) | Q. No: 6(a) | Marks: 3**

Comparing nuclear power with coal power.

**Batch: 16th Batch (2024) | Q. No: 6(b)(i) | Marks: 2**

Difference between: Nuclear fission and fusion

**Batch: 16th Batch (2024) | Q. No: 6(b)(ii) | Marks: 2**

Difference between: Renewable and nonrenewable resources.

**Batch: 16th Batch (2024) | Q. No: 7(c) | Marks: 3.5**

Write notes on: e-Waste

### Syllabus Topic: Ecosystem Ecology & Trophic Dynamics

**Batch: 16th Batch (2024) | Q. No: 5(a) | Marks: 3.5**

Describe about trophic categories according to how they get their energy.

**Batch: 16th Batch (2024) | Q. No: 5(b) | Marks: 3.5**

Explain resilience mechanisms that restored an ecosystem.

### Syllabus Topic: Natural Disasters & Management

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 3.5**

What is drought? Briefly describe different types of drought.

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 3.5**

Discuss about impacts of drought.

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 4**

Enlist natural disasters that encountered in Bangladesh. How do tsunamis and tornadoes affect the environment.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 3**

Discuss briefly the management plans for earthquakes and floods?

**Batch: 15th Batch (2023) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Desertification and human civilization

### Syllabus Topic: Human Population Dynamics & Urbanization

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 4.5**

Discuss the effect of population growth on our environment.

### Syllabus Topic: Environmental Impact & Risk Assessment

**Batch: 15th Batch (2023) | Q. No: 5(a) | Marks: 5**

With advantage and disadvantage explain the methods of environmental impact assessment.

**Batch: 15th Batch (2023) | Q. No: 5(b) | Marks: 2**

Define the four steps of environmental risk assessment.

**Batch: 16th Batch (2024) | Q. No: 2(c) | Marks: 2**

How to assess risk assessment.

# BOT4105: ELEMENTARY PLANT BREEDING

## Syllabus Topic: Introduction & History of Plant Breeding

**Batch: 11th Batch (2019) | Q. No: 1(a) | Marks: 2.5**

Briefly describe the contribution of different disciplines to plant breeding.

**Batch: 11th Batch (2019) | Q. No: 1(b) | Marks: 4.5**

Outline the history of plant breeding and mention its landmark achievements.

**Batch: 12th Batch (2020) | Q. No: 1(a) | Marks: 1**

Define plant breeding.

**Batch: 12th Batch (2020) | Q. No: 1(b) | Marks: 6**

Briefly describe the various objectives of plant breeding.

**Batch: 13th Batch (2021) | Q. No: 7(a) | Marks: 3.5**

Write notes on: History of plant breeding

**Batch: 14th Batch (2022) | Q. No: 1(a) | Marks: 1 | *REPEATED OCCURRENCE***

Define plant breeding.

**Batch: 14th Batch (2022) | Q. No: 1(b) | Marks: 6 | *REPEATED OCCURRENCE***

Briefly describe the objectives of plant breeding.

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 4 | *REPEATED OCCURRENCE***

Briefly describe the history of Plant Breeding.

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 3**

Describe the relationship between plant breeding and other branches of Biology.

**Batch: 15th Batch (2023) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Nature of plant breeding

**Batch: 16th Batch (2024) | Q. No: 1(a) | Marks: 2.0**

What are the objectives of plant breeding?

**Batch: 16th Batch (2024) | Q. No: 1(b) | Marks: 5.0**

Describe the role of plant breeding in improving crop yield, nutritional quality and resistance to different stresses.

## Syllabus Topic: Reproduction Methods & Gametogenesis

**Batch: 11th Batch (2019) | Q. No: 2(a) | Marks: 4.0**

Define sporogenesis. Briefly describe the process of micro and megasporogenesis with suitable diagrams.

**Batch: 11th Batch (2019) | Q. No: 2(b) | Marks: 3.0**

Define apomixis. Briefly describe different types of apomixis found in crop plants.

**Batch: 12th Batch (2020) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Asexual reproduction in crop plants

**Batch: 13th Batch (2021) | Q. No: 1(a) | Marks: 1.0**

Define reproduction?

**Batch: 13th Batch (2021) | Q. No: 1(b) | Marks: 4.0**

Briefly describe the asexual reproduction methods of crop plants.

**Batch: 13th Batch (2021) | Q. No: 1(c) | Marks: 2.0**

Write significance of asexual reproduction in plant breeding.

**Batch: 14th Batch (2022) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Sexual reproduction process of crop plants

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 1**

Define reproduction.

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 4**

Describe sexual reproduction method in crop plants.

**Batch: 15th Batch (2023) | Q. No: 2(c) | Marks: 2**

Write the significance of sexual reproduction in plant breeding.

**Batch: 16th Batch (2024) | Q. No: 2(a) | Marks: 3.0**

Define apomixis. How is it different from sexual reproduction? (Marks: 1.0 + 2.0)

**Batch: 16th Batch (2024) | Q. No: 2(b) | Marks: 4.0**

Explain different types of apomixis and its significance in maintaining hybrid vigor in crop plants. (Marks: 1.0 + 3.0)

**Batch: 16th Batch (2024) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Asexual reproduction methods in plants

## Syllabus Topic: Pollination & Fertilization Mechanisms

**Batch: 11th Batch (2019) | Q. No: 3(a) | Marks: 5.0**

Briefly describe the various mechanisms that promote self- and cross-pollination.

**Batch: 12th Batch (2020) | Q. No: 4(a)(i) | Marks: 2**

Differentiate the following pairs: Self and cross pollination

**Batch: 13th Batch (2021) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Mechanism which promote cross pollination

**Batch: 14th Batch (2022) | Q. No: 2(a) | Marks: 1**

Define pollination.

**Batch: 14th Batch (2022) | Q. No: 2(b) | Marks: 6**

Describe the mechanisms that promote self and cross pollination.

**Batch: 16th Batch (2024) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Mechanisms which promote self-pollination

## Syllabus Topic: Self-Incompatibility

**Batch: 11th Batch (2019) | Q. No: 3(b) | Marks: 2.0**

Briefly describe the various techniques for temporary suppression of self-incompatibility.

**Batch: 12th Batch (2020) | Q. No: 2(a) | Marks: 1**

What is self-incompatibility?

**Batch: 12th Batch (2020) | Q. No: 2(b) | Marks: 6**

Describe in brief the genetic basis of self-incompatibility.

**Batch: 14th Batch (2022) | Q. No: 3(a) | Marks: 1**
Define self-incompatibility.

**Batch: 14th Batch (2022) | Q. No: 3(b) | Marks: 6**
Explain the genetic basis of self-incompatibility.

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 2**
Briefly state different types of incompatibility found in plants.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 2**
Mention various methods used in overcoming self incompatibility.

**Batch: 15th Batch (2023) | Q. No: 3(c) | Marks: 3**
Explain heteromorphic self incompatibility.

### Syllabus Topic: Male Sterility

**Batch: 12th Batch (2020) | Q. No: 7(c) | Marks: 3.5**
Write notes on: Male sterility

**Batch: 13th Batch (2021) | Q. No: 2(a) | Marks: 5.0**
Briefly describe different types of male sterility in crop plants.

**Batch: 13th Batch (2021) | Q. No: 2(b) | Marks: 2.0**
Write significance of male sterility in plant breeding.

**Batch: 14th Batch (2022) | Q. No: 7(b) | Marks: 3.5**
Write notes on: Male sterility

**Batch: 16th Batch (2024) | Q. No: 3(a) | Marks: 1.0**
Define male sterility in plants.

**Batch: 16th Batch (2024) | Q. No: 3(b) | Marks: 4.0**
Describe the different types of male sterility.

**Batch: 16th Batch (2024) | Q. No: 3(c) | Marks: 2.0**
Why is male sterility an important tool in crop improvement?

### Syllabus Topic: Hybridization & Distant Hybridization

**Batch: 12th Batch (2020) | Q. No: 3(a) | Marks: 1**

Classify hybridization.

**Batch: 12th Batch (2020) | Q. No: 3(b) | Marks: 2**

Write the various objectives of hybridization.

**Batch: 12th Batch (2020) | Q. No: 3(c) | Marks: 4**

Briefly describe the techniques involved in artificial hybridization.

**Batch: 13th Batch (2021) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Emasculation

**Batch: 14th Batch (2022) | Q. No: 5(a) | Marks: 3**

Write the types and objectives of hybridization in plant breeding. (Marks: 1+2)

**Batch: 14th Batch (2022) | Q. No: 5(b) | Marks: 4**

Explain various techniques of artificial hybridization.

**Batch: 15th Batch (2023) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Technique of artificial hybridization

**Batch: 16th Batch (2024) | Q. No: 5(a) | Marks: 1.0**

Define hybridization in plant breeding.

**Batch: 16th Batch (2024) | Q. No: 5(b) | Marks: 4.0**

Describe the steps involved in artificial hybridization in plants.

**Batch: 16th Batch (2024) | Q. No: 5(c) | Marks: 2.0**

Why is hybridization important in plant breeding?

## Syllabus Topic: Selection Methods in Self-Pollinated Crops

**Batch: 11th Batch (2019) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Merits and demerits of mass selection

**Batch: 12th Batch (2020) | Q. No: 4(a)(ii) | Marks: 2**

Differentiate the following pairs: Pureline and mass selection.

**Batch: 12th Batch (2020) | Q. No: 4(b) | Marks: 3**

Describe in short the field techniques of pureline selection.

**Batch: 13th Batch (2021) | Q. No: 3(a) | Marks: 1.0**

Define mass selection.

**Batch: 13th Batch (2021) | Q. No: 3(b) | Marks: 4.0**

Briefly describe the method of mass selection.

**Batch: 13th Batch (2021) | Q. No: 3(c) | Marks: 2.0**

Write merits and demerits of mass selection.

**Batch: 14th Batch (2022) | Q. No: 4(a) | Marks: 4**

Write the differences between mass selection and pure-line selection.

**Batch: 14th Batch (2022) | Q. No: 4(b) | Marks: 3**

Briefly describe the application of pure-line selection.

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 1**

What is mass selection?

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 4**

Describe the procedure of mass selection.

**Batch: 15th Batch (2023) | Q. No: 4(c) | Marks: 2**

Write merits and demerits of mass selection in crop improvement.

**Batch: 16th Batch (2024) | Q. No: 4(a) | Marks: 5.0**

Briefly describe different types of selection methods employed for crop improvement.

**Batch: 16th Batch (2024) | Q. No: 4(b) | Marks: 2.0**

Write the differences between mass and pure line selections.

## Syllabus Topic: Plant Introduction & Acclimatization

**Batch: 11th Batch (2019) | Q. No: 4(a) | Marks: 1.5**

Define plant introduction.

**Batch: 11th Batch (2019) | Q. No: 4(b) | Marks: 2.0**

Describe in brief the objectives of plant introduction.

**Batch: 11th Batch (2019) | Q. No: 4(c) | Marks: 3.5**

Describe the role of plant introduction in Bangladesh.

**Batch: 12th Batch (2020) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Role of plant introduction in Bangladesh

**Batch: 13th Batch (2021) | Q. No: 4(a) | Marks: 1.0**

What do you mean by plant introduction?

**Batch: 13th Batch (2021) | Q. No: 4(b) | Marks: 2.0**

Explain the purpose of plant introduction.

**Batch: 13th Batch (2021) | Q. No: 4(c) | Marks: 4.0**

Write merits and demerits of plant introduction.

**Batch: 14th Batch (2022) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Merits and demerits of plant introduction

**Batch: 15th Batch (2023) | Q. No: 5(a) | Marks: 1**

Define plant introduction.

**Batch: 15th Batch (2023) | Q. No: 5(b) | Marks: 6**

Describe the procedures of plant introduction.

## Syllabus Topic: Origin & Domestication of Crops

**Batch: 11th Batch (2019) | Q. No: 5 | Marks: 7.0**

Briefly describe the centres of origin proposed by N.I. Vavilov.

**Batch: 11th Batch (2019) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Center of origin of rice

**Batch: 12th Batch (2020) | Q. No: 5(a) | Marks: 1**

Define domestication.

**Batch: 12th Batch (2020) | Q. No: 5(b) | Marks: 6**

Briefly describe the various important changes that have occurred in crop plants under domestication.

**Batch: 13th Batch (2021) | Q. No: 5(a) | Marks: 5.0**

Briefly describe the main centers of origin of crop species.

**Batch: 13th Batch (2021) | Q. No: 5(b) | Marks: 2.0**

Write the significance of the concept of origin of crop in plant breeding.

**Batch: 14th Batch (2022) | Q. No: 6(a) | Marks: 4**

Briefly describe the criticisms of Vavilov's centre of origin concept.

**Batch: 14th Batch (2022) | Q. No: 6(b) | Marks: 3**

Write the changes of crop plants under domestication.

**Batch: 15th Batch (2023) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Changes under domestication

**Batch: 16th Batch (2024) | Q. No: 6(a) | Marks: 1.0**

What is meant by the 'centre of origin' of crop plants?

**Batch: 16th Batch (2024) | Q. No: 6(b) | Marks: 6.0**

Describe Vavilov's eight centres of origin of crop plants.

## Syllabus Topic: Plant Genetic Resources & Germplasm Conservation

**Batch: 11th Batch (2019) | Q. No: 6(a) | Marks: 2.0**

Define germplasm.

**Batch: 11th Batch (2019) | Q. No: 6(b) | Marks: 5.0**

Briefly describe the conservation strategies of germplasm.

**Batch: 12th Batch (2020) | Q. No: 6(a) | Marks: 1**

What do you mean by plant genetic resources?

**Batch: 12th Batch (2020) | Q. No: 6(b) | Marks: 4**

Briefly describe the collection strategies of germplasm.

**Batch: 12th Batch (2020) | Q. No: 6(c) | Marks: 2**

Describe in brief the uses of germplasm in plant breeding program.

**Batch: 13th Batch (2021) | Q. No: 6(a) | Marks: 1.0 | *REPEATED OCCURRENCE***

Define germplasm.

**Batch: 13th Batch (2021) | Q. No: 6(b) | Marks: 6.0 | *REPEATED OCCURRENCE***

Briefly describe the conservation methods of germplasm.

**Batch: 15th Batch (2023) | Q. No: 6(a) | Marks: 1 | *REPEATED OCCURRENCE***

What is plant genetic resources?

**Batch: 15th Batch (2023) | Q. No: 6(b) | Marks: 3**

Write down the different types and sources of plant genetic resources.

**Batch: 15th Batch (2023) | Q. No: 6(c) | Marks: 3**

Briefly describe the ex-situ conservation of plant genetic resources.

**Batch: 16th Batch (2024) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Ex-situ conservation of germplasm

# BOT4106: BIOSTATISTICS

## Syllabus Topic: t-Test & Significance

**Batch: 11th Batch (2019) | Q. No: 1(a) | Marks: 2.0**

What is t test? Write the application of t test.

**Batch: 11th Batch (2019) | Q. No: 1(b) | Marks: 5.0**

Using the following data perform a significance test to compare the means of strain A and B and comment on the result obtained: Strain A: $\Sigma x = 279$ mm, $n = 10$, $\Sigma x^2 = 8852.0$ Strain B: $\Sigma x = 271$ mm, $n = 8$, $\Sigma x^2 = 9480.0$

**Batch: 12th Batch (2020) | Q. No: 1(a) | Marks: 2**

Define 't' test. Write the characteristics of paired 't' test.

**Batch: 12th Batch (2020) | Q. No: 1(b) | Marks: 5**

Two lentil varieties (var. A and B) were cultivated in ten pairs of equal sized plot. Yield/plot (Kg) was given in the following table. Comment on significance of differences between two means of yield:

| Plot Pair | var. A Yield (Kg) | var. B Yield (Kg) |
|---|---|---|
| 1 | 9.5 | 10.0 |
| 2 | 12.0 | 9.5 |
| 3 | 13.5 | 8.5 |
| 4 | 11.0 | 10.0 |
| 5 | 9.5 | 10.5 |
| 6 | 12.5 | 8.5 |
| 7 | 14.0 | 8.5 |
| 8 | 11.5 | 9.0 |
| 9 | 12.4 | 10.8 |
| 10 | 11.9 | 8.5 |

**Batch: 13th Batch (2021) | Q. No: 1(a) | Marks: 2**

Explain 't' test.

**Batch: 13th Batch (2021) | Q. No: 1(b) | Marks: 5**

Protein content found in maize grains of two varieties A & B are given in the following table. Determine whether the difference in protein content is significant:

| Variety | Protein Content (gm) |
|---|---|
| Protein in variety A | 15.1, 14.1, 14.5, 15, 12, 14, 13 |
| Protein in variety B | 12.5, 12, 10, 16, 9, 11, 12 |

**Batch: 14th Batch (2022) | Q. No: 1(a) | Marks: 2 | *REPEATED OCCURRENCE***

What is t test? Write the application of t test.

**Batch: 14th Batch (2022) | Q. No: 1(b) | Marks: 5**

Yield/plot (kg) in ten irrigated and ten non-irrigated rice plots are given below. Determine the significance of irrigation on rice production with comments using t-test:

| Plots | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Irrigated (kg) | 12.0 | 10.5 | 11.0 | 13.0 | 12.0 | 8.5 | 13.0 | 14.5 | 15.5 | 12.0 |
| Non-irrigated (kg) | 10.0 | 9.0 | 8.5 | 11.0 | 9.5 | 9.0 | 8.0 | 10.5 | 7.0 | 8.0 |

**Batch: 15th Batch (2023) | Q. No: 1(a) | Marks: 1 | *REPEATED OCCURRENCE***

Write the characteristics of paired 't' test.

**Batch: 15th Batch (2023) | Q. No: 1(b) | Marks: 6**

Biomass production in reserved area and non reserved area in each of eight quadrats was as follows. Determine the significance of difference in two means of biomass production areas:

| Quadrat | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Reserved ($gm/m^2$) | 48 | 50 | 38 | 45 | 49 | 52 | 35 | 45 |
| Non-reserved ($gm/m^2$) | 40 | 35 | 30 | 35 | 38 | 42 | 36 | 25 |

**Batch: 16th Batch (2024) | Q. No: 1(a) | Marks: 1 | *REPEATED OCCURRENCE***

Define the t-test.

**Batch: 16th Batch (2024) | Q. No: 1(b) | Marks: 1 | *REPEATED OCCURRENCE***

Write the application of the t-test.

**Batch: 16th Batch (2024) | Q. No: 1(c) | Marks: 5**

Protein content found in 100 gm of rice of two varieties A and B are given in the following table. Determine whether the difference in protein content is significant:

| Variety | Protein content (gm) |
| --- | --- |
| Variety A (gm) | 2.6, 1.9, 3.0, 2.1, 1.6, 2.2, 1.8 |
| Variety B (gm) | 2.0, 1.8, 2.2, 1.5, 1.9, 2.5, 2.3 |

## Syllabus Topic: Correlation

**Batch: 11th Batch (2019) | Q. No: 2(a) | Marks: 2.0**

What are correlation and coefficient of correlation?

**Batch: 11th Batch (2019) | Q. No: 2(b) | Marks: 5.0**

Describe different types of correlation with figures.

**Batch: 12th Batch (2020) | Q. No: 2(a) | Marks: 1**

What is Co-efficient of correlation?

**Batch: 12th Batch (2020) | Q. No: 2(b) | Marks: 6**

Plant weight (gm/plant) and grain weight (gm/plant) in seven observations are given below. Determine the co-efficient of correlation and draw proper comment:

| Obs. No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| --- | --- | --- | --- | --- | --- | --- | --- |
| Plant Weight (gm) | 12.5 | 10.0 | 7.5 | 8.5 | 11.5 | 9.5 | 8.0 |
| Grain Weight (gm) | 6.5 | 8.0 | 6.5 | 7.0 | 7.5 | 8.0 | 7.0 |

**Batch: 12th Batch (2020) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Types of correlation

**Batch: 13th Batch (2021) | Q. No: 2(a) | Marks: 2 | *REPEATED OCCURRENCE***

Define correlation and co-efficient of correlation.

**Batch: 13th Batch (2021) | Q. No: 2(b) | Marks: 5 | *REPEATED OCCURRENCE***

Explain different types of correlation with figures.

**Batch: 14th Batch (2022) | Q. No: 2(a) | Marks: 1 | *REPEATED OCCURRENCE***

What is correlation?

**Batch: 14th Batch (2022) | Q. No: 2(b) | Marks: 6**

Using the following data on plant height and yield of six wheat plants, find out the correlation of these two traits:

| Plant Height (cm) | Grain Yield (gm) |
|---|---|
| 30 | 8 |
| 36 | 9 |
| 28 | 7 |
| 40 | 10 |
| 32 | 10 |
| 24 | 8 |

**Batch: 15th Batch (2023) | Q. No: 2(a) | Marks: 2 | *REPEATED OCCURRENCE***

What is correlation and co-efficient of correlation?

**Batch: 15th Batch (2023) | Q. No: 2(b) | Marks: 5 | *REPEATED OCCURRENCE***

Describe different types of correlation with figures.

**Batch: 16th Batch (2024) | Q. No: 2(a) | Marks: 2 | *REPEATED OCCURRENCE***

Define correlation and the coefficient of correlation.

**Batch: 16th Batch (2024) | Q. No: 2(b) | Marks: 5**

Soil pH and plant height from seven plants are given below. Calculate the coefficient of correlation of the two variables:

| Soil pH | Plant Height (cm) |
|---|---|
| 4.5 | 15 |
| 5.0 | 16 |
| 5.5 | 17 |
| 6.0 | 18 |
| 6.5 | 18.5 |
| 7.0 | 19.5 |
| 7.5 | 19.5 |

## Syllabus Topic: Regression

**Batch: 11th Batch (2019) | Q. No: 3(a) | Marks: 2.0**

Define regression and regression coefficient.

**Batch: 11th Batch (2019) | Q. No: 3(b) | Marks: 5.0**

Nitrogen fertilizers were applied at six levels on a wheat variety to test the yield performance of grains. Determine the regression coefficient and regression equation using the following data:

| N level (kg/h) | 0 | 20 | 40 | 60 | 80 | 100 |
|---|---|---|---|---|---|---|
| Yield (t/h) | 3.2 | 4.25 | 4.75 | 5.50 | 6.0 | 6.30 |

**Batch: 12th Batch (2020) | Q. No: 3(a) | Marks: 1**

Define regression.

**Batch: 12th Batch (2020) | Q. No: 3(b) | Marks: 1.5**

What do you mean by dependent character and independent character?

**Batch: 12th Batch (2020) | Q. No: 3(c) | Marks: 4.5**

Increase of fungal colony growth (cm) with the increase of media pH is given in the following table. Determine the regression coefficient and regression equation:

| Media pH | Colony growth (cm) |
|---|---|
| 4.5 | 7 |
| 5.0 | 17 |
| 5.5 | 30 |
| 6.0 | 50 |
| 6.5 | 80 |
| 7.0 | 85 |

**Batch: 13th Batch (2021) | Q. No: 3(a) | Marks: 1 | *REPEATED OCCURRENCE***

Define regression.

**Batch: 13th Batch (2021) | Q. No: 3(b) | Marks: 6**

Urea dose and yield in a rice variety are given below. Determine regression and regression equation:

| Urea Dose (kg/plot) | Yield (kg/plot) |
|---|---|
| 0 | 9 |
| 3 | 10.5 |
| 6 | 12 |
| 9 | 14 |
| 12 | 15.5 |
| 15 | 16 |

**Batch: 14th Batch (2022) | Q. No: 3(a) | Marks: 2**

Define regression and linear regression.

**Batch: 14th Batch (2022) | Q. No: 3(b) | Marks: 3**

Nitrogen fertilizer was applied at different doses for increasing yield of a rice variety. Determine the regression coefficient and regression equation from the given data:

| Nitrogen doses (kg/plot) | Yield (kg) |
|---|---|
| 0 | 9.5 |
| 2 | 11 |
| 4 | 14 |
| 6 | 16 |
| 8 | 18 |
| 10 | 18.5 |

**Batch: 15th Batch (2023) | Q. No: 3(a) | Marks: 1 | *REPEATED OCCURRENCE***

Define regression.

**Batch: 15th Batch (2023) | Q. No: 3(b) | Marks: 6**

Ear length (cm) and seed weight (gm) of six wheat plants are given below. Determine the regression co-efficient and regression equation:

| Ear length (cm) | Seed weight (gm) |
|---|---|
| 7.5 | 22 |
| 6.0 | 20 |
| 8.0 | 25 |
| 9.0 | 30 |
| 10 | 35 |
| 8.5 | 30 |

**Batch: 16th Batch (2024) | Q. No: 3(a) | Marks: 1 | *REPEATED OCCURRENCE***

Define regression.

**Batch: 16th Batch (2024) | Q. No: 3(b) | Marks: 6**

The weight and height of seven individuals are as follows. Find out the regression and the regression equation:

| Weight (lb) | Height (inch) |
|---|---|
| 135 | 61 |
| 140 | 63 |
| 145 | 68 |
| 150 | 69 |
| 155 | 70 |
| 160 | 71 |
| 165 | 73 |

## Syllabus Topic: Principles of Experimental Design

**Batch: 11th Batch (2019) | Q. No: 4(a) | Marks: 3.0**

Define following terms: experiment, treatment, yield and block.

**Batch: 11th Batch (2019) | Q. No: 5(a) | Marks: 3**

Explain the principles of experimental design.

**Batch: 12th Batch (2020) | Q. No: 4(a) | Marks: 4.0**

Define the following terms: (i) Experiment, (ii) experimental design, (iii) treatment, and (iv) yield.

**Batch: 12th Batch (2020) | Q. No: 4(b) | Marks: 2.0**

Describe one factor experiment and two factors experiment.

**Batch: 12th Batch (2020) | Q. No: 5(a) | Marks: 3.0**

Describe the principles of experimental design.

**Batch: 13th Batch (2021) | Q. No: 4(a) | Marks: 2.5**

Explain one factor and two factor experiment.

**Batch: 14th Batch (2022) | Q. No: 4(a) | Marks: 1**

What are single factor and double factor experiments?

**Batch: 14th Batch (2022) | Q. No: 4(b) | Marks: 3**

Explain the basic principles of experimental design.

**Batch: 14th Batch (2022) | Q. No: 4(c) | Marks: 3**

Define the following terms: Experiment, treatment, and yield.

**Batch: 15th Batch (2023) | Q. No: 4(a) | Marks: 3**

With examples explain one way and two way analysis of variances.

**Batch: 15th Batch (2023) | Q. No: 4(b) | Marks: 4**

Define the following terms: i) Experiment; ii) Experimental design; iii) Treatment; iv) Yield.

**Batch: 16th Batch (2024) | Q. No: 4(a) | Marks: 2**

Differentiate single factor experiment and two factor experiment.

**Batch: 16th Batch (2024) | Q. No: 4(b) | Marks: 2**

What do you mean by experiment and experimental design?

**Batch: 16th Batch (2024) | Q. No: 4(c) | Marks: 3**

Explain the basic principles of an experimental design.

## Syllabus Topic: CRD, RCBD, and LSD Designs

**Batch: 11th Batch (2019) | Q. No: 4(b) | Marks: 2.0**

What are the source of variation in (i) CRD, (ii) RCBD and (iii) LSD.

**Batch: 11th Batch (2019) | Q. No: 4(c) | Marks: 2.0**

Write the characteristics of Randomized Complete Block Design (RCBD).

**Batch: 11th Batch (2019) | Q. No: 5(b) | Marks: 2**

Prepare a layout for Latin Square Design with four treatments.

**Batch: 11th Batch (2019) | Q. No: 5(c) | Marks: 2**

What are the advantages and disadvantages of Latin Square Design?

**Batch: 11th Batch (2019) | Q. No: 6(a) | Marks: 7.0**

An experiment was conducted in a Completely Randomized Design (CRD) at a green house to test the yield performance of five potato varieties A, B, C, D, and E. The yield (Kg/plot) is given below. Analyze the data, prepare an ANOVA table, and comment on significance of treatments:

| Row 1 | Row 2 | Row 3 | Row 4 |
|---|---|---|---|
| A 119 | C 200 | D 125 | B 121 |
| E 210 | B 123 | E 191 | A 150 |
| C 180 | A 150 | D 120 | C 165 |
| B 126 | E 190 | C 150 | B 125 |
| E 220 | D 117 | A 100 | D 130 |

**Batch: 12th Batch (2020) | Q. No: 4(c) | Marks: 1**

What are the sources of variation in (i) RCBD and (ii) LSD.

**Batch: 12th Batch (2020) | Q. No: 5(b) | Marks: 2+2**

What are the characteristics of Latin Square Design? Write down a comparative statement for CRBD and Latin Square Design.

**Batch: 12th Batch (2020) | Q. No: 6 | Marks: 7**

An experiment was conducted in a Randomized Complete Block Design to find out the yield performance of five varieties of Wheat (A, B, C, D and E). Carry out the analysis of variance and draw conclusion on yield performance of varieties:

| Variety | Block I | Block II | Block III | Block IV |
|---|---|---|---|---|
| A | 18.5 | 16.0 | 23.0 | 21.0 |
| B | 16.0 | 19.5 | 14.0 | 17.0 |
| C | 13.5 | 10.5 | 15.0 | 16.0 |
| D | 11.0 | 12.0 | 9.5 | 10.5 |
| E | 12.0 | 15.0 | 13.5 | 11.0 |

**Batch: 13th Batch (2021) | Q. No: 4(b) | Marks: 4.5**

What are the source of variation in CRD, RCBD and LSD.

**Batch: 14th Batch (2022) | Q. No: 5(a) | Marks: 2**

Write the characteristics of Completely Randomized Design.

**Batch: 14th Batch (2022) | Q. No: 5(b) | Marks: 2**

Mention the merits and limitations of Latin Square Design.

**Batch: 14th Batch (2022) | Q. No: 5(c) | Marks: 3**

What are the sources of variation in CRD, RCBD and LSD.

**Batch: 14th Batch (2022) | Q. No: 6 | Marks: 7**

In an experiment of RCB design yield performance of five wheat varieties (A, B, C, D, and E) are given in the following table. Carry out the analysis of variance and make proper comments:

| Variety | Block I | Block II | Block III |
|---|---|---|---|
| A | 10.5 | 8.0 | 6.5 |
| B | 15.5 | 20.0 | 12.5 |
| C | 9.5 | 12.0 | 7.5 |
| D | 12.0 | 7.5 | 9.0 |
| E | 14.0 | 8.5 | 11.0 |

**Batch: 15th Batch (2023) | Q. No: 5(a) | Marks: 2+2**

What are the characteristics of Randomized Complete Block design and Latin Square design?

**Batch: 15th Batch (2023) | Q. No: 5(b) | Marks: 3**

Mention the source of variation in CRD, RCBD and LSD.

**Batch: 16th Batch (2024) | Q. No: 5 | Marks: 7**

Discuss the differences between CRD, RCBD and LSD in terms of their applications, sources of variation, advantages and disadvantages.

## Syllabus Topic: Data Transformations & Multiple Comparisons

**Batch: 11th Batch (2019) | Q. No: 7(a) | Marks: 3.5**

Write notes on: DMRT

**Batch: 11th Batch (2019) | Q. No: 7(b) | Marks: 3.5**

Write notes on: Square root transformation

**Batch: 11th Batch (2019) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Least significance difference

**Batch: 12th Batch (2020) | Q. No: 7(a) | Marks: 3.5**

Write notes on: Arcsine transformation

**Batch: 12th Batch (2020) | Q. No: 7(b) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write notes on: DMRT

**Batch: 14th Batch (2022) | Q. No: 7(a) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write notes on: DMRT

**Batch: 14th Batch (2022) | Q. No: 7(b) | Marks: 3.5 | *REPEATED OCCURRENCE***

Write notes on: Least significant difference

**Batch: 14th Batch (2022) | Q. No: 7(c) | Marks: 3.5**

Write notes on: Logarithmic transformation